

Supplementary Material

Coordinate-Explicit and Route-Specific Finite-Precision Qualification of Tangent-Only Perturbations in Coupled Finite-Element Linearizations

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Purpose, version, and scope

This supplement accompanies manuscript v0.13.0. It preserves the registered analytical checks, exact finite-element operand provenance, route-specific finite-precision error components, response-route validation, independent precision audit, effectivity anatomy, tooling lineage, and claim boundaries. The M8 and M16 cases are independent fixed finite-element linearizations, not a mesh sequence. G2-20 added synthetic response-route evaluation and a newly generated holdout, but it did not execute Abaqus, modify a finite-element operator, or import the scalable sparse-estimator results reserved for Paper 3.

1 Analytical verification

Table S1: Registered reproduction of the directional non-determination construction

ε	μ_J (matched)	ρ_1 (matched)	ρ_g (matched)	aligned η_z	orthogonal η_z
1e-12	1.0000e-12	1.0000e-12	1.0000e-12	1.0000e-12	0
1e-10	1.0000e-10	1.0000e-10	1.0000e-10	1.0000e-10	0
1e-8	1.0000e-08	1.0000e-08	1.0000e-08	1.0000e-08	0
1e-6	1.0000e-06	1.0000e-06	1.0000e-06	1.0000e-06	0
1e-4	1.0000e-04	1.0000e-04	9.9990e-05	9.9990e-05	0
1e-2	1.0000e-02	1.0000e-02	9.9010e-03	9.9010e-03	0

Table S2: Registered coordinate-transformation and redeclaration calculation

s	transformed $\widehat{\Delta J}_{22}$	transformed physical $(\Delta J)_{22}$	transformed η_z	redeclared physical $(\Delta J)_{22}$	redeclared η_z
1	1.0000e-06	1.0000e-06	1.0000e-06	1.0000e-06	1.0000e-06
10	1.0000e-05	1.0000e-06	1.0000e-06	1.0000e-07	1.0000e-07
1e3	1.0000e-03	1.0000e-06	1.0000e-06	1.0000e-09	1.0000e-09
1e6	1.0000e+00	1.0000e-06	1.0000e-06	1.0000e-12	1.0000e-12

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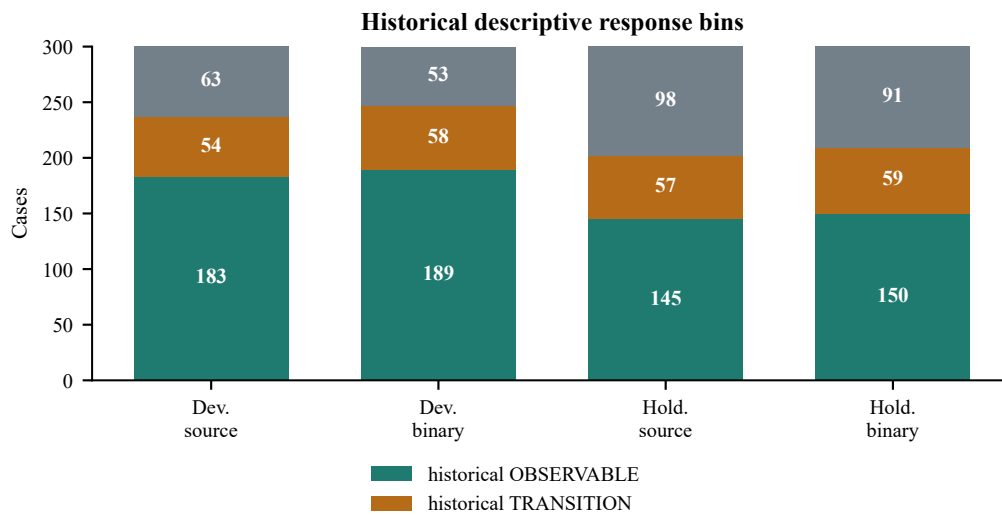
Table S3: Historical reproduction and direct-route coverage for the two reference problems

Dataset/lane	Coverage	Severe undercoverage	Evidence role
historical development	287/300 (13 uncovered)	3	reproduced motivation
development source	300/300 (0 uncovered)	0	pre-specified coverage
development binary	300/300 (0 uncovered)	0	pre-specified coverage
holdout source	300/300 (0 uncovered)	0	pre-specified coverage
holdout binary	300/300 (0 uncovered)	0	pre-specified coverage



The two source-representation markers are registered components and are not summed. This explains the historical truth-contract mismatch; it does not attribute a guard defect.

Figure S1: Source-representation attribution for the three historical severe-undercoverage cases. This retained diagnostic does not assign a perturbation defect



Diagnostic labels only: no validated class boundary and are excluded from confirmatory conclusions.

Figure S2: Historical descriptive response bins retained for provenance only. The labels have no validated class boundary and are excluded from confirmatory conclusions

2 Response-route theorem validation

For the response equation

$$A\delta z = -Ez, \quad A = J + E,$$

the computed route uses

$$\tilde{f} = \text{fl}(-E\tilde{z}), \quad r_\delta = \tilde{f} - A\tilde{\delta z}, \quad q_f = \tilde{f} + E\tilde{z}.$$

With majorants $\beta_A \geq \|A^{-1}\|_2$, $\alpha_E \geq \|E\|_2$, and $\|\tilde{z} - z\|_2 \leq U_z$, the implemented bound is

$$U_r = \beta_A \alpha_E U_z + \beta_A \|q_f\|_2 + \beta_A \|r_\delta\|_2.$$

The development design was reused only after this route model was fixed. A new 300-system holdout then used disjoint seeds and registered condition, dimension, perturbation-amplitude, and family grids.

Table S4: Direct- and response-route coverage and lower-quantity counts

Set	Reference contract	Direct coverage	Response coverage	Lower response error	Lower response enclosure
Development	Decimal source	300/300	300/300	295/300	283/300
Development	Binary64 operand	300/300	300/300	199/300	229/300
New holdout	Decimal source	300/300	300/300	271/300	229/300
New holdout	Binary64 operand	300/300	300/300	160/300	188/300

All 2,400 route–contract–split coverage decisions passed: 300/300 for both routes in both reference contracts on development and on the new holdout. The response equation had lower decimal-source reference error in 295/300 development and 271/300 new-holdout systems. Binary64 counts were 199/300 and 160/300. These data do not establish a universal route ranking.

3 Precision and implementation independence

The registered high-precision implementation was repeated at 120 and 180 decimal digits. The maximum normalized difference was 5.16×10^{-110} on development and 6.82×10^{-107} on the new holdout, with no coverage-decision change. A separately implemented Python `Decimal` backend evaluated 36 stratified holdout systems in both reference contracts. Across 72 lane comparisons, the maximum normalized difference from the registered reference was 6.96×10^{-153} .

The first G2-20 execution stopped because its precision comparator normalized near-zero quantities by their own magnitude. That attempt remains classified as an inconclusive reference-or-precision result. Two tooling-only corrections replaced the comparator without changing operators, thresholds, response formulas, or scientific outputs; the corrected run restarted from row 1. This lineage is reported rather than overwritten.

4 Effectivity anatomy and extreme ratios

For positive reference error e_{ref} and response magnitude d_{ref} , define

$$I_{\text{eff}} = \frac{U}{e_{\text{ref}}}, \quad R_U = \frac{U}{d_{\text{ref}}}.$$

For a parent bound $U = \|A^{-1}\|_F \|r\|_2$ and exact forward error $e = \|A^{-1}r\|_2$,

$$I_{\text{eff}} = \underbrace{\frac{\|A^{-1}\|_F}{\|A^{-1}\|_2}}_{\text{norm-majorant inflation}} \underbrace{\frac{\|A^{-1}\|_2 \|r\|_2}{\|A^{-1}r\|_2}}_{\text{directional inflation}}.$$

Table S5: Factorized direct-route effectivity anatomy

Set	Quantity	Median range	95th-percentile range
Development	Frobenius majorization	1.0001–1.0001	1.94–1.94
Development	Directional inflation	1.18×10^4 – 2.31×10^4	3.18×10^8 – 8.53×10^8
Original holdout	Frobenius majorization	1.0050–1.0050	2.69–2.69
Original holdout	Directional inflation	1.36×10^3 – 1.88×10^3	3.59×10^7 – 8.72×10^7

The norm-majorant median was approximately 1.0001 on development and 1.0050 on the original holdout. Directional-inflation medians instead ranged from 1.36×10^3 to 2.31×10^4 . Thus the central conservativeness is associated primarily with replacing the realized residual direction by a worst-direction inverse norm.

The largest finite direct-route effectivity was 7.9321×10^{86} , reported in the manuscript as $\log_{10} I_{\text{eff}} = 86.90$. It occurred for `HOLDOUT_SPD_D16_C01_E3eM17`, whose reference error was approximately 4.9995×10^{-102} . The extreme ratio is therefore a near-zero-denominator event, not undercoverage. Exact strings and case identifiers remain in `Paper2_extreme_ratio_record.json`.

A registered main-effects analysis gave adjusted $R^2 = 0.869$ on development; the drop-one partial R^2 for condition exponent was 0.855. The original holdout fit gave adjusted $R^2 = 0.057$ and condition partial $R^2 = 0.029$. These finite-grid results support mechanism analysis within the registered design, not a universal conditioning law.

5 Frozen finite-element operand provenance

Case	Dimension	Snapshot SHA-256	Declared operand contract
M8	196	ACF00D823AAE 8BC073F14ECF A322AED1B004 CBBA85D5FA83 416FEC4D24D87482	Frozen IEEE binary64 J, J_g, b , free map, and scales; transformed free coordinates; Euclidean norm.
M16	900	22A51CA5BB44 03D1EB9499F4 588D6EF19CE7 D6C3636429D5 EE2FCA6AE3AB207E	Independently frozen IEEE binary64 J, J_g, b , free map, and scales; transformed free coordinates; Euclidean norm.

The M8 and M16 matrix snapshots were reproduced bitwise by independent capture witnesses. Each application reconstructed every declared binary64 operand exactly before high-precision comparison. The identity $A = J + E$ was checked elementwise before either response route was evaluated.

Table S6: Rounded finite-element route values; exact decimal strings remain in the machine-readable decision files

Case	Route	d_{obs}	U	L	U/d_{ref}
M8	Direct	2.919e−12	5.880e−13	2.330e−12	2.523e−1
M8	Response Eq.	2.920e−12	2.588e−22	2.920e−12	8.864e−11
M16	Direct	2.463e−11	3.971e−12	2.066e−11	1.922e−1
M16	Response Eq.	2.464e−11	6.930e−21	2.464e−11	2.813e−10

6 Finite-element route results

7 Finite-element precision, residual, identity, and repeatability

Table S7: Rounded precision and identity checks; all gates passed

Case	120/180 difference	Maximum residual	Route-identity difference	Repeat difference
M8	8.81×10^{-121}	2.55×10^{-124}	9.18×10^{-123}	0
M16	3.97×10^{-121}	4.14×10^{-124}	3.04×10^{-122}	0

M8 required 468 s and 0.11 GiB recorded peak memory. M16 required 10.4 h and 1.35 GiB. These are offline reference costs for two fixed applications; they are not a sparse-complexity fit or an online-estimator benchmark.

8 Finite-element route-specific error components

Table S8: Component-level coverage and final route enclosure

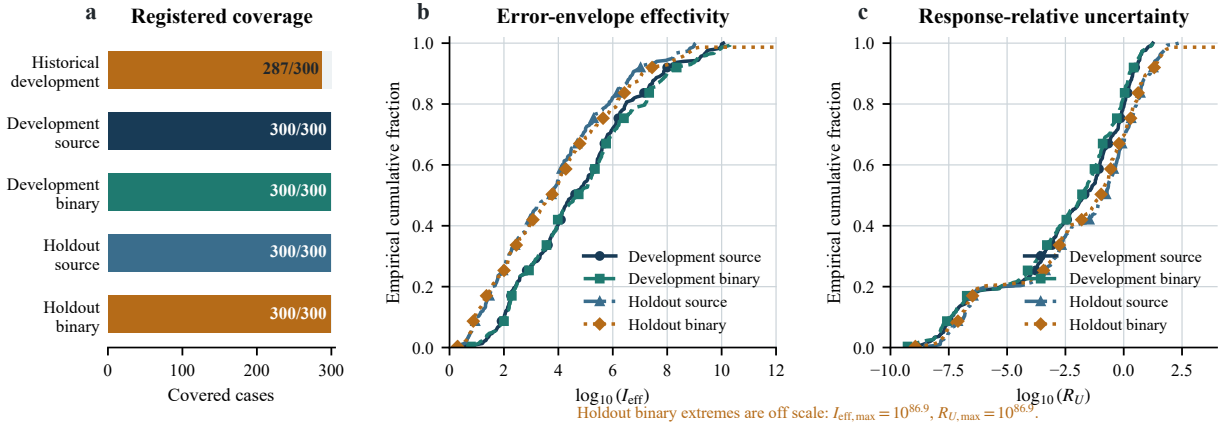
Case	Route	Registered components	Final U	Coverage
M8	Direct difference	two parent solves; subtraction	5.880×10^{-13}	3/3
M8	Response equation	parent propagation; forcing; response solve	2.588×10^{-22}	3/3
M16	Direct difference	two parent solves; subtraction	3.971×10^{-12}	3/3
M16	Response equation	parent propagation; forcing; response solve	6.930×10^{-21}	3/3

All registered components covered their independently evaluated reference errors for the two fixed applications. Coverage is operation-graph and case specific; exact component strings remain in the archived ledgers.

9 Historical diagnostics retained for provenance

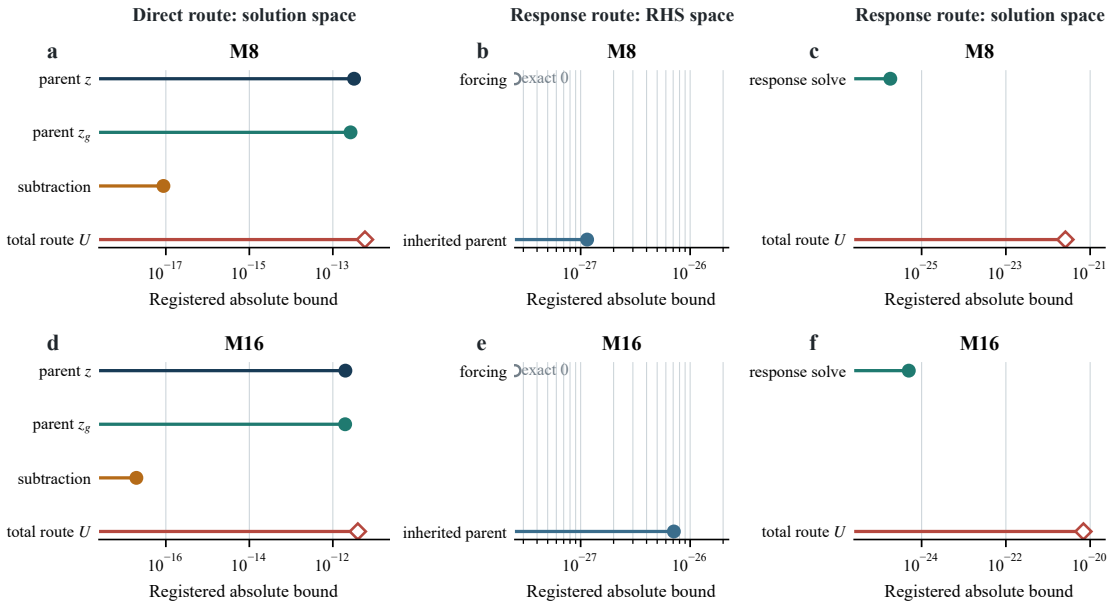
10 Claim boundaries and machine-readable ledger

The CSV file `Paper2_claim_evidence_matrix_v0.10.0.csv` remains the authoritative record for claims frozen through v0.10.0. The G2-20 execution report and adjudication extend it for the response-route theorem, new holdout, independent backend, and effectivity analysis; historical claim IDs and evidence tiers were not rewritten. The complete row-level matrix is distributed as CSV rather than typeset at an illegible scale in this PDF. Table S9 retains the concise inference boundary needed for human review.



Complete coverage does not imply a tight or computationally efficient uncertainty enclosure.

Figure S3: Historical coverage and aggregate conservativeness retained for provenance. The main manuscript replaces this summary with factorized effectivity anatomy



Component spaces are separated and are not summed across panels. The decomposition explains M8 and M16 only; it does not rank routes universally.

Figure S4: Historical route-component anatomy for M8 and M16. Right-hand-side and solution-space components are not added across spaces

Table S9: Interpretation boundary by claim domain

Domain	Established	Not established
Theory	Finite response identity, directional non-determination, and coordinate mapping under declared assumptions.	Prevalence, optimal scaling, or a universal guard rule.
Synthetic evidence	Reproduction, coverage, and effectivity on the specified development and disjoint holdout designs.	Unrestricted-system coverage, calibrated tightness, or computational efficiency.
Fixed tangents	Route-specific intervals completed separately for M8 and M16.	Mesh law, dimension trend, or population-level generality.
Route comparison	Each route has its own uncertainty components and a case-specific decision.	Universal route ranking or superiority.
Interpretation	Relation of $[L, H]$ to an optional threshold supplied independently by the problem definition; zero exclusion when $\tau = 0$.	Universal threshold selection, guard safety or harm, nonlinear consequence, finite-element stability, or engineering qualification.
Cost	Recorded offline cost and resource provenance for each frozen case.	Online estimator feasibility, complexity fit, or production scaling.

Data and code availability

The existing non-confidential reproducibility artifact is archived as Zenodo concept DOI 10.5281/zenodo.21833850; the public implementation is available from the GitHub repository. The v1.1.0 G2-20 release package contains the new holdout, route results, independent-backend audit, analysis tables, scripts, and hashes. The archive excludes Abaqus executables, user-element source, solver state histories, and engineering project data.